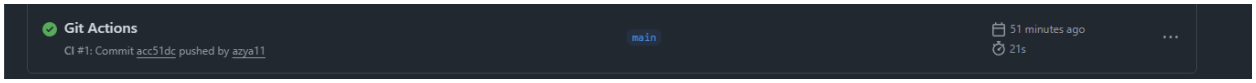
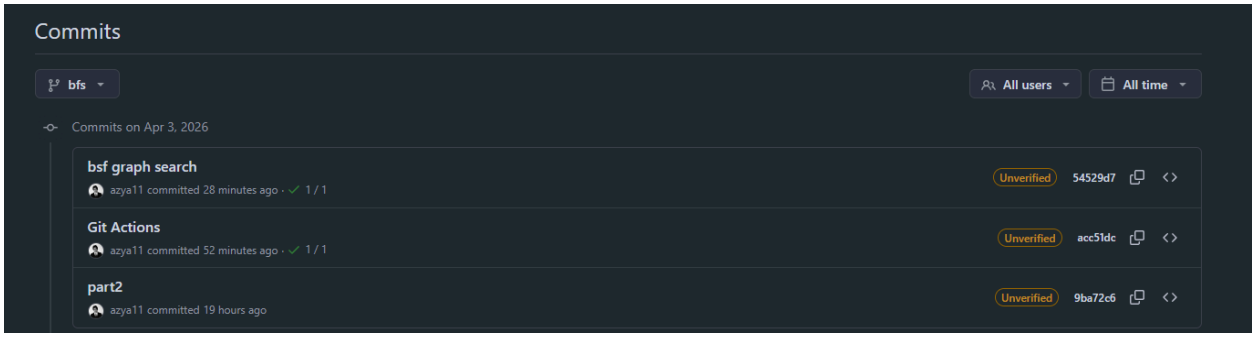
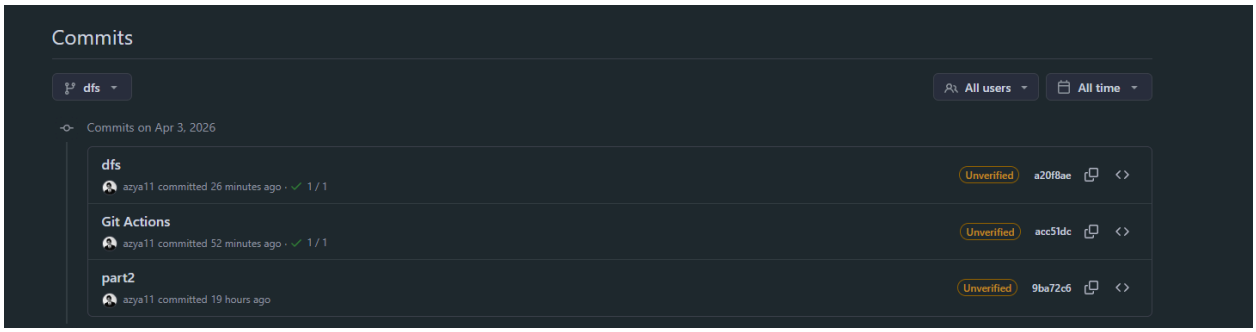
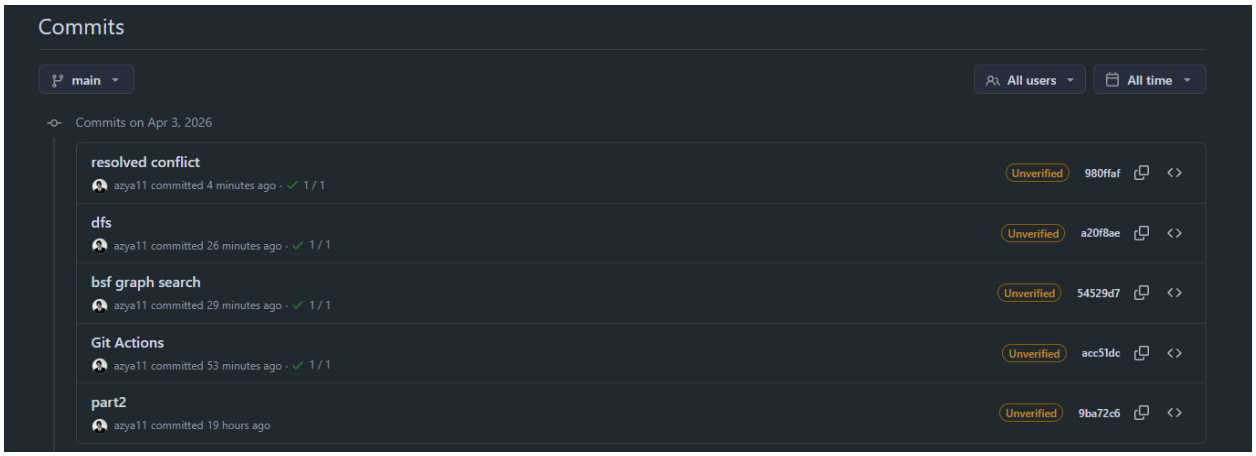
https://github.com/azya11/CSE4642026-ashamura-asu.edu
https://github.com/azya11/CSE4642026-ashamura-asu.edu/tree/dfs
https://github.com/azya11/CSE4642026-ashamura-asu.edu/tree/bfs









```
PS C:\Users\azizf\Desktop\CSE4642026-ashamura-asu.edu> mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< asu.cse464:CSE4642026-azizf >--------------------
[INFO] Building CSE4642026-azizf 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE4642026-azizf ---
[INFO] skip non existing resourceDirectory C:\Users\azizf\Desktop\CSE4642026-ashamura-asu.edu\src\main
\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE4642026-azizf ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE4642026-azizf ---
[INFO] skip non existing resourceDirectory C:\Users\azizf\Desktop\CSE4642026-ashamura-asu.edu\src\test
\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE4642026-azizf ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ CSE4642026-azizf ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running graph.GraphTest
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.111 s -- in graph.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ CSE4642026-azizf ---
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.193 s
[INFO] Finished at: 2026-04-03T21:17:30-07:00
[INFO] ------------------------------------------------------------------------
PS C:\Users\azizf\Desktop\CSE4642026-ashamura-asu.edu>
```

```
=== Part 1: Remove APIs ===
Before removal:
Number of nodes: 4
Nodes: [a, b, c, d]
Number of edges: 3
Edges:
a -> b
b -> c
c -> d

After removeEdge(c, d):
Number of nodes: 4
Nodes: [a, b, c, d]
Number of edges: 2
Edges:
a -> b
b -> c

After removeNode(b):
Number of nodes: 3
Nodes: [a, c, d]
Number of edges: 0
Edges:

After removeNodes([a, c]):
Number of nodes: 1
Nodes: [d]
Number of edges: 0
Edges:

Exception caught: Node not found: z
Exception caught: Edge not found: x->y
```

```
=== Part 5: GraphSearch (BFS & DFS) ===
Graph:
Number of nodes: 5
Nodes: [a, b, c, d, e]
Number of edges: 5
Edges:
a -> b
a -> c
b -> d
c -> d
d -> e

BFS path from a to e: a -> b -> d -> e
DFS path from a to e: a -> b -> d -> e
BFS path from a to z (no path): null
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.023 s
[INFO] Finished at: 2026-04-03T21:20:08-07:00
[INFO] ------------------------------------------------------------------------
PS C:\Users\azizf\Desktop\CSE4642026-ashamura-asu.edu>
```